

POWERED BY **ZAIZENS** · AN INTEGRATED HMS

PREMIUM · PROFESSIONAL EDITION

Maternity & Antenatal Care (ANC)

Complete staff guide — ANC registry, obstetric chart, labor ward, WHO partograph, delivery, newborn registration, postnatal care, and billing.

PRODUCT**ZAIZENS Integrated HMS****DOCUMENT TYPE****Maternity User Manual****WEBSITE****<https://www.zaizens.com/>****ORGANISATION****TSSF (Shisong Annex) SOA**Version **2.0**Generated **14 June 2026 at 18:44**Platform **Node.js HMS Edition**Languages **English · Français**

Introduction

Welcome to the **ZAIZENS Integrated HMS Maternity User Manual**. This document covers the full obstetric pathway — from first ANC booking through antenatal visits, labor monitoring, delivery, newborn care, and postnatal follow-up.

Module scope: Obstetrics & gynaecology — ANC registry, maternity charts, labor ward, partograph, delivery documentation, neonatal linkage, and maternity billing tariffs.

9

CHART TABS

3

RISK LEVELS

98+

MATERNITY SERVICES



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1. Maternity Overview & Roles

The **Maternity module** manages antenatal, intrapartum, and postnatal care with integrated billing, IPD admission, and newborn registration.

**Obstetrician / Doctor**
ANC reviews, risk scoring, scans, labor management, delivery, complications

**Midwife / Nurse**
ANC vitals, partograph entries, delivery assistance, postnatal checks

**Front Desk**
Link HMS patient, book ANC, route to cashier for MAT tariffs

**Cashier**
ANC booking fees, delivery charges, postnatal visits — MAT payment codes

**Lab / Ultrasound**
ANC labs (Hb, HIV, blood group), anomaly scans, Doppler

**ADT / Ward**
Labor ward admission, maternity bed assignment, NICU pediatric beds

Key URLs

Screen	Path	Roles
Maternity dashboard	/maternity	All maternity staff
ANC registry	/maternity/patients	Doctor, Nurse, Front Desk
ANC booking	/maternity/register	Front Desk, Nurse
Maternity chart	/maternity/chart/:id	Doctor, Midwife
Labor ward	/maternity/labor	Midwife, Doctor
Patient chart tab	Clinical chart → Maternity	All clinical
Death registry	/maternity/death-registry	Doctor, Admin

Permissions:

maternity.read

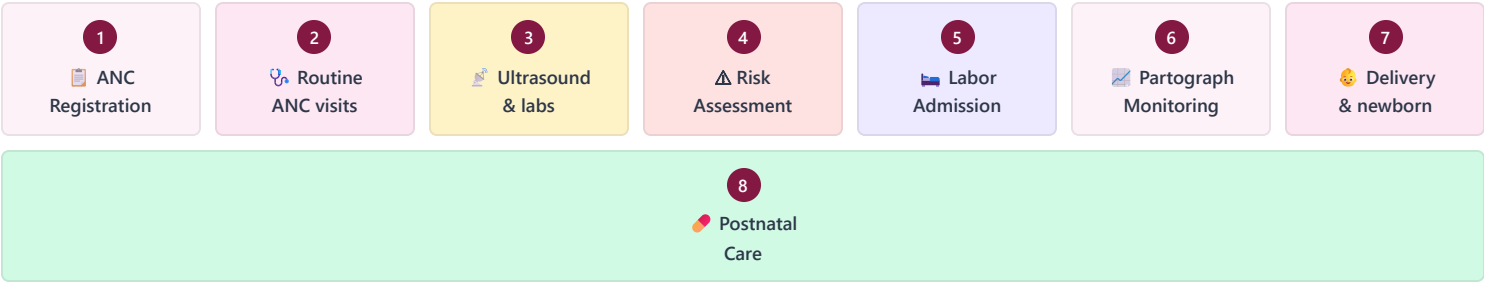
maternity.write

clinical.write

nursing.write

2. Maternity Care Pathway

The standard obstetric journey through the HMS maternity module:



ANC Registration (G/P/A, LMP, EDD) → Routine ANC visits (vitals, FHR, Hb)
→ Ultrasound & risk scoring → Admit to labor ward → WHO partograph
→ Record delivery → Register newborn → Postnatal visits → [NICU if required]

Integration with other modules

- **OPD** — pediatric assessment for newborn; follow-up consultations
- **IPD** — labor ward admission, maternity beds, NICU pediatric ward
- **Emergency** — emergency labor with pay-later deposit option
- **Cashier** — MAT module payment tickets and maternity service catalog

3. Dashboard & Navigation

The **Maternity dashboard** at `/maternity` shows:

- **Registered** — total maternity patients
- **Active ANC** — ongoing pregnancies
- **In labor** — active labor ward patients
- **Deliveries today** — births recorded today

Quick links

Link	Purpose
Registry	Full ANC patient list with filters
New ANC booking	Register new pregnancy
Labor ward	Active labors and partograph
Death registry	Maternal/neonatal mortality

High-risk panel

Dashboard highlights **high-risk ANC** patients (risk score $\geq$ 20) sorted by EDD for proactive management.

4. ANC Registration

Book new ANC

- 1 Open `/maternity/register` or **Book ANC** from dashboard.
- 2 Link existing HMS patient (Patient directory → `?from=maternity`) or enter Patient ID.
- 3 Enter obstetric history: **Gravida, Para, Abortions**.
- 4 Record **LMP** — system calculates **EDD** (Naegele rule: LMP + 280 days).
- 5 Blood group, HIV status, PMTCT enrollment if applicable.
- 6 Save — system assigns **ANC number** (e.g. ANC-2026-0001).
- 7 Redirect to maternity chart for first visit.

First visit: ANC booking visit tariff applies automatically. Collect payment at cashier or defer per hospital policy.

5. Maternity Chart

Each pregnancy has a dedicated chart at `/maternity/chart/:id` with tabs:

ANC visits

Risk

Ultrasound

Labor

Delivery

Newborn

Postnatal

Complications

Chart header

Displays ANC number, patient name, LMP, EDD, EGA (weeks), and risk level badge (LOW MEDIUM HIGH).

From the chart: admit to labor, record delivery, register newborn, open linked IPD admission, collect MAT payments, and document complications.

6. ANC Visits

Record routine ANC visit

- 1 Open chart → **ANC visits** tab.
- 2 Enter visit date — EGA auto-calculated from LMP.
- 3 Record: weight, BP (sys/dia), fetal heart rate (FHR), Hb, presentation.
- 4 Schedule next visit date; add clinical notes.
- 5 Save visit — appears in visit history table.

Visit history

Table shows date, EGA, BP, FHR, Hb for trend monitoring across pregnancy.

Clinical integration

ANC vitals sync to patient clinical chart. Maternity episode visible on patient chart Maternity tab.

7. Risk Assessment

WHO-style simplified risk scoring on the **Risk** tab:

Risk factors (examples)

- Age under 18 or over 35
- Grand multipara, previous caesarean section
- Previous PPH or stillbirth
- Hypertension, diabetes, severe anemia
- HIV positive (PMTCT pathway)
- Multiple pregnancy, malpresentation, placenta previa
- Pre-eclampsia, inadequate ANC attendance

Scoring

Score	Level	Action
< 10	LOW	Routine ANC schedule
10–19	MEDIUM	Increased monitoring, specialist review
≥ 20	HIGH	Dashboard alert, delivery planning, hospital birth

Document action plan and save assessment. Review at each ANC visit.

8. Ultrasound Scans

Record scans on the **Ultrasound** tab:

- **Scan types:** Dating, Anomaly (20–24 weeks), Growth, Doppler
- Findings, placenta location, amniotic fluid index (AFI)
- Anomaly detected flag for referral pathway

Scan history table tracks all obstetric ultrasounds with type and anomaly status.

9. Labor Ward & Partograph

STEP 1 — ADMIT TO LABOR WARD

- 1 Chart → **Labor** tab when patient in active labor.
- 2 Enter admission type, EGA, cervical dilation (cm).
- 3 Option: **Create IPD admission** (Maternity Ward) — auto-assign bed when available.
- 4 **Pay later** — defer HOS payment for emergency labor.
- 5 Submit **Admit & open partograph**.

STEP 2 — WHO PARTOGRAPH MONITORING

At `/maternity/labor` , select active labor from sidebar:

- Record time, dilation, station, contractions (/10 min), FHR, BP
- System flags **alert line** and **action line** crossings
- Timeline table shows all partograph entries

Active labors list on dashboard and labor ward page for shift handover.

10. Delivery & Newborn Registration

STEP 2 — RECORD DELIVERY

- 1 From chart **Delivery** tab or labor ward → Record delivery.
- 2 Delivery type: SVD, vacuum, forceps, emergency/elective CS, breech.
- 3 Outcome, estimated blood loss (ml), date/time.
- 4 Save delivery record — triggers billing catalog mapping.

STEP 3 — REGISTER NEWBORN

- 1 **Newborn** tab — sex, birth weight, APGAR 1 and 5 min.
- 2 Birth outcome: alive, fresh stillbirth.
- 3 Birth immunizations: Vit K, BCG, Hep B; breastfeeding initiated.
- 4 Create HMS patient record for baby; link to mother chart.
- 5 Option: **NICU admission** (Pediatric Ward IPD) with pay-later for prematurity/RDS.
- 6 **Pediatric OPD** assessment for well newborn follow-up.

11. Postnatal Care

Postnatal tab documents:

- Routine postnatal visits (Day 1, Day 7, Week 6)
- Maternal recovery assessment
- Infant feeding and umbilical care
- Contraception counseling
- MAT tariff: Postnatal Visit — Routine

Update maternity patient status to **Delivered** when episode complete.

12. Complications

Document on **Complications** tab:

Complication	Documentation
PPH	Postpartum hemorrhage — blood loss, interventions
Pre-eclampsia / eclampsia	Severity, magnesium, delivery timing
Sepsis	Source, antibiotics, ICU transfer
Obstructed labor	Partograph action line, CS decision

Link to **Death registry** for maternal or neonatal mortality.

13. Billing & MAT Tickets

Maternity module integrates with Service Catalog maternity tariff (98+ services):

- **ANC booking** — first visit
- **ANC routine** — follow-up visits
- **Labor ward admission**
- **Delivery types** — SVD, vacuum, forceps, CS (emergency/elective)
- **Postnatal routine**
- **NICU per day**
- Neonatal items: BCG, OPV, incubator, phototherapy, kangaroo care

Chart billing panel

Shows paid/due status, suggested tariffs for recorded events, MAT payment tickets on file. Route patient to cashier with payment code.

14. NICU & Pediatric Linkage

For infants requiring intensive care:

- 1 After delivery registration, choose **NICU admission**.
- 2 System creates pediatric IPD admission on available NICU/pediatric bed.
- 3 Neonatal catalog charges (incubator, phototherapy) bill to baby patient.
- 4 Mother and baby charts cross-linked on patient records.
- 5 Emergency labor may use **pay later** for both mother IPD and baby NICU deposits.

15. Screen Reference (UI)

Maternity Dashboard — /maternity

Obstetrics & gynaecology

142
ACTIVE ANC

3
IN LABOR

5
DELIVERIES TODAY

High-risk ANC

ANC-2026-0042 · A. Ngu · EDD 15 Jul · HIGH

Maternity Chart — ANC-2026-0018

LMP 01 Sep 2025 · EDD 08 Jun 2026 · EGA 32w · MEDIUM

ANC

Risk

Scans

Labor

Delivery

Newborn

Date	EGA	BP	FHR	Hb
14 Jun	32w	118/72	142	11.2

Labor Ward — Partograph

Active labor · Jane Doe · dilation 6 cm

Time	Dil.	Station	CTX	FHR	BP
1430	4	-2	3/10	148	120/70
1530	6	-1	4/10	140	118/72

Live screenshots: Open `/maternity` while signed in; Print → Save as PDF for training materials.

16. Troubleshooting

Issue	Solution
Cannot register ANC	Load HMS patient first; check maternity.write permission
EDD incorrect	Verify LMP date; ultrasound dating scan may override
High-risk not on dashboard	Save risk assessment with score ≥ 20 ; status must be active
Cannot admit labor	Use Labor tab on chart; complete admission form with dilation
Partograph empty	Select patient from Active labors sidebar on /maternity/labor
Delivery blocked	Admit to labor ward first (Step 1 before Step 2)
Newborn not linked	Complete delivery record before newborn registration
MAT ticket missing	Record clinical event; check billing panel for suggested tariff
NICU bed unavailable	Manual bed assignment on ward board; pay-later if configured

17. Glossary

Term	Definition
ANC	Antenatal Care — pregnancy monitoring before birth
G/P/A	Gravida / Para / Abortions — obstetric history
LMP	Last Menstrual Period — start of pregnancy dating
EDD	Expected Date of Delivery (Naegele rule)
EGA	Estimated Gestational Age (weeks)
FHR	Fetal Heart Rate
Partograph	WHO labor monitoring chart (dilation, contractions, FHR)
PMTCT	Prevention of Mother-to-Child Transmission (HIV)
PPH	Postpartum Hemorrhage
SVD	Spontaneous Vaginal Delivery
CS	Caesarean Section
APGAR	Newborn vitality score at 1 and 5 minutes
NICU	Neonatal Intensive Care Unit